

FastAPI exam

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ wget https://dst-de.s3.eu-west-3.amazonaws.com/fastapi_en/questions_en.xlsx
--2025-06-23 09:07:50-- https://dst-de.s3.eu-west-3.amazonaws.com/fastapi_en/questions_en.xlsx
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 52.95.155.40, 3.5.224.12
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|52.95.155.40|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11954 (12K) [application/vnd.openxmlformats-officedocument.spreadsheetml.sheet]
Saving to: 'questions_en.xlsx'

questions_en.xlsx      100%[=====] 11.67K  --KB/s   in 0s

2025-06-23 09:07:50 (106 MB/s) - 'questions_en.xlsx' saved [11954/11954]
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ pip install -r requirements.txt
Requirement already satisfied: fastapi in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from -r requirements.txt (line 1)) (0.115.12)
Requirement already satisfied: uvicorn[standard] in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from -r requirements.txt (line 2)) (0.33.0)
Collecting pandas
  Downloading pandas-2.0.3-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (12.4 MB)
    | 12.4 MB 15.0 MB/s
Collecting openpyxl
  Downloading openpyxl-3.1.5-py3-none-any.whl (250 kB)
    | 250 kB 38.9 MB/s
Requirement already satisfied: typing-extensions>=4.8.0 in /home/ubuntu/datascientest-data-engineering/pyspark/.ve
```

After completion of downloads and installed packages, run the following command in the terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api/exam']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [3068] using WatchFiles
INFO: Started server process [3070]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open new terminal and run the following:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ chmod +x tests.sh
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ ./tests.sh
HTTP/1.1 200 OK
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 15
content-type: application/json

{"status":"ok"}HTTP/1.1 401 Unauthorized
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 56
content-type: application/json

{"detail":"Missing or malformed `Authorization` header"}HTTP/1.1 422 Unprocessable Entity
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 38
content-type: application/json

{"detail":"Count must be 5, 10 or 20"}HTTP/1.1 404 Not Found
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 31
content-type: application/json

{"detail":"No questions match"}HTTP/1.1 403 Forbidden
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 24
content-type: application/json

{"detail":"Admins only"}HTTP/1.1 200 OK
date: Mon, 23 Jun 2025 12:05:40 GMT
server: uvicorn
content-length: 121
content-type: application/json

{"question":"What?","subject":"Geo","use":"MCQ","correct":["C"],"answerA":"A","answerB":"B","answerC"}
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$
```

Open URL <http://127.0.0.1:8000/docs> and execute to test the ok status:

MCQ Service 1.0 OAS 3.1

/openapi.json

Serve randomised quiz questions

utilities

GET

/health

Health check

questions

GET

/questions

Get a batch of random questions

POST

/questions

Create a new question (admin only)

Schemas

HTTPValidationError > Expand all object

Question > Expand all object

ValidationError > Expand all object

utilities

GET

/health

Health check

Parameters

No parameters

Execute Clear

Responses

Curl

curl -X 'GET' \ 'http://127.0.0.1:8000/health' \ -H 'accept: application/json'

Request URL

http://127.0.0.1:8000/health

Server response

Code

Details

200

Response body

{ "status": "ok" }

Response headers

content-length: 15 content-type: application/json date: Mon, 23 Jun 2025 12:07:56 GMT server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

Open URL <http://127.0.0.1:8000/health>

```
{  "status": "ok"}
```

RawParsed

Open URL <http://127.0.0.1:8000/docs> again and try to execute another example: questions

GET

/questions

Get a batch of random questions

Cancel

Name

Description

use * required

string

(query)

Test type (e.g. MCQ)

MCQ

subject * required

array<string>

(query)

One or more subjects

Add item

-

Add string item

count * required

integer

(query)

Number of questions (5, 10, 20)

20

maximum: 20

minimum: 1

Authorization

string | (string | null)

(header)

Basic alice:wonderland

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/questions?use=MCQ&subject=Add%20item&count=20' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/questions?use=MCQ&subject=Add%20item&count=20
```

Server response		
Code	Details	
401	Error: Unauthorized Response body <pre>{ "detail": "Missing or malformed 'Authorization' header" }</pre> Download Response headers <pre>content-length: 56 content-type: application/json date: Mon, 23 Jun 2025 12:23:33 GMT server: uvicorn</pre>	
Responses		
Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>[{ "question": "string", "subject": "string", "use": "string", "correct": ["string"], "answerA": "string", "answerB": "string", "answerC": "string", "answerD": "string" }]</pre>	No links
401	Unauthorised	No links
404	No questions found	No links
422	Validation error	No links

Open Request URL

<http://127.0.0.1:8000/questions?use=MCQ&subject=Add%20item&count=20>

```
{
  "detail": "Missing or malformed `Authorization` header"
}
```

Raw

Parsed

```
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● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/exam$ ./tests.sh
HTTP/1.1 200 OK
date: Mon, 23 Jun 2025 13:19:12 GMT
server: uvicorn
content-length: 15
content-type: application/json

{"status":"ok"}HTTP/1.1 401 Unauthorized
date: Mon, 23 Jun 2025 13:19:12 GMT
server: uvicorn
content-length: 56
content-type: application/json

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{"detail":"No questions match"}HTTP/1.1 403 Forbidden
date: Mon, 23 Jun 2025 13:19:12 GMT
server: uvicorn
content-length: 24
content-type: application/json

{"detail":"Admins only"}HTTP/1.1 201 Created
date: Mon, 23 Jun 2025 13:19:12 GMT
server: uvicorn
content-length: 121
content-type: application/json

{"question":"What?","subject":"Geo","use":"MCQ","correct":["C"],"answerA":"A","answerB":"B","answerC":"C",
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